COURSE 2 · SECTION 10 · EXERCISE

DFMEA risk before and after

Section 10 · DFMEA risk view

Start: Section11_Start.pbix or your Engxtech_S10.pbix; save a copy as Engxtech_S10_Exercise.pbix

Time: about 15 min

Data: the course data folder

DESIGN QUESTION

Which design risks were reduced by the recommended actions, which ones are still high, and which high-severity failure modes have no action at all?

Extra practice. This exercise is extra practice: a longer task than the video's short "Your turn" questions, which are answered in the video itself. Do it after the video, in a copy of your file.

DFMEA has one row per failure mode with Severity (S), Occurrence (O) and Detection (D) before and after the action. RPN = S x O x D. Severity normally stays the same unless the design removes the effect. The video added the columns RPN Before, RPN After and HighS NoAction, and the measures Total RPN Before and Total RPN After.

Your tasks

1. In your exercise copy, read DF-01 in the before/after column chart, and the Total RPN Before and Total RPN After cards.
2. New measures in _Measures: **Lines RPN Before >= 100** and **Lines RPN After >= 100**.
3. Read the 'high S, no action' table (filter HighS NoAction = True). Add O, D and Action to it.
4. S x O matrix (rows S, columns O, Count of LineID): how many lines have S = 9?
5. Read DF-02 and DF-08: does the type of action fit the score that moved?

Check your answer

Check	Expected
DF-01 RPN before / after	**280 / 72**
Sum of RPN before / after	**1470 / 788**
Lines with RPN of 100 or more, before / after	**8 / 2**
High severity (S at least 7) with no action	**DF-04, DF-07**
Lines with S = 9	**3**

Hint. The two lines still at RPN 100 or more after the actions are exactly the two high-severity lines with no action.

Stuck? The worked solution with the full DAX and M is in Solutions/S10_Solution.pdf.